No.: 00000

(Testing) - Harris's New Form API4000 DRAFT5

Date: 21/07/2026

INSTRUMENT IDENTIFICATION

INNOVATIVE LAB SERVICES LLC - API 4000 LC/MS/MS SYSTEM - PLANNED MAINTENANCE PROCEDURE (Doc 1_1, Rev 1.1)

Instrument Serial #	Eyeye
Instrument Name / INV #	Test

(Testing) - Harris's New Form API4000 DRAFT5

No.: 00000
Date: 21/07/2026

CUSTOMER INFORMATION

Customer Site	Nestle
Serial Number	2646464
Customer Name	Nextle
Service Engineer	Harris
Date	07/23/2026

SIGN OFF

By signing this form you acknowledge that the planned maintenance was completely performed.

Customer

Service Engineer

PRE-PM 1.0 VACUUM PRESSURE CHECK

Guideline: CAD = 0 -> <= 0.8x10-5 torr (0.08 V). CAD = 12 -> <= 5.5x10-5 torr (+/- 0.3x10-5).

Vacuum pressure, CAD = 0	.25
Vacuum pressure, CAD = 12	

PRE-PM 2.0 POSITIVE ION MODE TEST (10 SCANS MCA)

Guideline (10 Scans MCA): Q1 m/z 906.673 >= 2.0x10^7 | Q1 m/z 2242.6 >= 1.0x10^6 | Q3 m/z 906.673 >= 2.0x10^7 | Q3 m/z 2242.6 >= 8.0x10^5. FWHM 0.6 - 0.8 amu throughout.

Q1 m/z 906.673

Intensity	
FWHM	

Q1 m/z 2242.6

Intensity	
FWHM	

Q3 m/z 906.673

Intensity	
FWHM	

Q3 m/z 2242.6

Intensity	
FWHM	

PRE-PM 3.0 NEGATIVE ION MODE TEST (10 SCANS MCA)

Guideline (10 Scans MCA): Q1 m/z 933.636 >= 1.1x10^7 | Q1 m/z 2037.4 >= 1.0x10^6 | Q3 m/z 933.636 >= 9.0x10^6. FWHM 0.6 - 0.8 amu throughout. Q3 m/z 2037.4 is N/A.

Q1 m/z 933.636

Intensity	
FWHM	

Q1 m/z 2037.4

Intensity	
FWHM	

Q3 m/z 933.636

Intensity	
FWHM	

PRE-PM 4.0 RESERPINE MS/MS TEST (10 SCANS MCA)

Guideline (10 Scans MCA): MS/MS transmission efficiency (m/z 195.0 / attenuated m/z 609.3) x 100 >= 10.0 %

Attenuated m/z 609.3

Intensity	
FWHM	0.70

m/z 195.0

Intensity	
FWHM	
Transmission Efficiency %	

PM SECTION - MAINTENANCE AND CLEANING

1. Check operation of the Turbo V ion source heaters	☑
2. Check lens voltages (Lens Power Supply module)	☑
3. Check RF tuning voltages (Q1 and Q3 QPS AMP module). Only if required, tune QPS AMPs.	☑
4. Check operation of the instrument cooling fans	☑
CEM voltage (V)	

SHUTDOWN AND PERFORM MAINTENANCE / CLEANING

AC input voltage (V)	
7. Inspect gas tubing for breakage (CAD gas, interface gas tubes, etc)	☐
8. Remove and clean the curtain plate	☐
9. If necessary, remove and clean the orifice plate	☐
10. If necessary, clean the front end of Q0	☐
11. Replace the air filter	☐
12. Replace the roughing pumps oil	☐
13. Inspect the oil return systems and if necessary replace the oil filter	☐
14. If applicable, perform PM on Peak Scientific gas generator	☐
15. Reinstall all covers and pump down	☑
16. Re-check operation of the instrument cooling fans	☑
17. If necessary, perform Turbo V ion source maintenance	☑
18. Clear computer temp files (Start / Search / Files and Folders -> *.tmp -> Local Hard Drive (C), (D) -> Search now -> Edit / Select All / Files / Delete / Empty Recycle Bin)	☐

PERFORMANCE VERIFICATION 1.0 GAS PRESSURE

Low Gas 1 / Gas 2 pressure can affect turbo spray and sensitivity. Source Exhaust pressure above 60 psi can cause exhaust valve failures.

Requirement: Gas 1 / Gas 2 100 - 105 psi | Curtain / CAD Gas 60 psi | Source Exhaust Supply 55 - 60 psi

Gas 1 / Gas 2 (psi)	
Curtain / CAD Gas (psi)	60.0
Source Exhaust Supply (psi)	

PERFORMANCE VERIFICATION 2.0 VACUUM PRESSURE

Guideline: CAD = 0 -> <= 0.8x10-5 torr (0.08 V). CAD = 12 -> <= 5.5x10-5 torr (+/- 0.3x10-5).

Vacuum pressure, CAD = 0	
Vacuum pressure, CAD = 12	

PERFORMANCE VERIFICATION 3.0 POSITIVE ION MODE TEST (10 SCANS MCA)

Guideline (10 Scans MCA): Q1 m/z 906.673 >= 2.0x10^7 | Q1 m/z 2242.6 >= 1.0x10^6 | Q3 m/z 906.673 >= 2.0x10^7 | Q3 m/z 2242.6 >= 8.0x10^5. FWHM 0.6 - 0.8 amu throughout.

Q1 m/z 906.673

Intensity	
FWHM	0.60

Q1 m/z 2242.6

Intensity	
FWHM	

Q3 m/z 906.673

Intensity	
FWHM	

Q3 m/z 2242.6

Intensity	
FWHM	

PERFORMANCE VERIFICATION 4.0 NEGATIVE ION MODE TEST (10 SCANS MCA)

Guideline (10 Scans MCA): Q1 m/z 933.636 >= 1.1x10^7 | Q1 m/z 2037.4 >= 1.0x10^6 | Q3 m/z 933.636 >= 9.0x10^6. FWHM 0.6 - 0.8 amu throughout. Q3 m/z 2037.4 is N/A.

Q1 m/z 933.636

Intensity	
FWHM	

Q1 m/z 2037.4

Intensity	
FWHM	

Q3 m/z 933.636

Intensity	
FWHM	

PERFORMANCE VERIFICATION 5.0 RESERPINE MS/MS TEST (10 SCANS MCA)

Guideline (10 Scans MCA): MS/MS transmission efficiency (m/z 195.0 / attenuated m/z 609.3) x 100 >= 10.0 %

Attenuated m/z 609.3

Intensity	
FWHM	

m/z 195.0

Intensity	
FWHM	
Transmission Efficiency %	

6.0 TEST SPECTRA AND COMMENTS

Test spectra printed and attached	☐
Comments or Observations	This is a test
Annotation	

This is a test